Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **SQL**, **Python**, and **Azure** data platforms, **ETL/ELT** pipelines, **Power BI** reporting, and LLM-enabled solutions integrated with enterprise data.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Designed normalized multi-tenant schemas on **Azure SQL** and SQL Server with row-level security for 500+ users, cutting dashboard latency from 1.2s to 350ms through **query tuning** and indexing optimization.
- Built **ETL/ELT** pipelines with Python, FastAPI, and SQL transformations across PostgreSQL, SQL Server, and Blob Storage, running 5K+ monthly **incremental loads** with reconciliation checks at 99%+ production reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Integrated **Claude** and GPT-4o multimodal LLMs with OCR into a **Python**/Flask grading platform for 150+ instructors, handling 10K+ daily grading requests and cutting average feedback turnaround time by 60%.
- Built concurrent batch ingestion for PDF, DOCX, and ZIP files over **MySQL** on AWS RDS, processing 1K+ files per batch while tracking per-call token usage and spend for **cost monitoring**.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Redesigned **ETL** workflows with AWS Glue and S3, transforming raw JSON, CSV, and Parquet logs into normalized analytics-ready datasets, boosting pipeline efficiency 68% and cutting storage costs 15%.
- Optimized large-scale API integrations with **SQL indexing**, caching, pagination, and field filtering, improving response times 30% and automating Jenkins CI/CD, unit, and integration tests across three operating systems.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built eight microservices with Node.js, React, REST APIs, and WebSockets over **Azure SQL**, modeling users, resources, invoices, and leads while synchronizing 2M+ records across locations in real time.
- Delivered **Power BI** dashboards over Azure SQL usage and allocation data, monitored with Azure Monitor, guiding operational decisions that reduced resource wastage 42% and raised lead conversions 15%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built hourly **Azure Databricks** PySpark pipelines orchestrated by **Azure Data Factory** over 50K+ booking and payment events, aggregating peak hours, service demand, and revenue for Power BI reporting.
- Developed Flask booking, pricing, coupon, and payment APIs with Firebase real-time sync, sustaining sub-200ms **API latency** and automating idempotent webhook refund handling, cutting manual payment handling by 70%.

Skills

Languages: Python, SQL, T-SQL, PySpark, Spark SQL, DAX, Java, TypeScript, JavaScript, Shell
Data Platforms: Power BI, Azure Databricks, Azure Data Factory, Snowflake, SQL Server, Azure SQL, PostgreSQL, MySQL
Data Engineering: ETL/ELT, Data Modeling, Star Schema, Data Warehousing, Incremental Loads, Data Validation
Performance & Governance: Query Optimization, Indexing, Performance Tuning, Row-Level Security, Reconciliation
Cloud & AI: Microsoft Azure, Azure Blob Storage, Azure Monitor, AWS (S3, Glue, RDS), Claude, OpenAI GPT-4o, RAG
DevOps & Practices: CI/CD, Docker, Git, GitHub Actions, Jenkins, Agile/Scrum, Stakeholder Collaboration

Projects

Project Controls Lakehouse | *Python, PySpark, Azure Data Factory, Databricks, Snowflake, Power BI, DAX*

- Designed a medallion lakehouse moving cost-code and labor-hour data through Azure Data Factory and Databricks into a Snowflake star schema powering an incrementally refreshed **Power BI semantic model** with **DAX** earned-value measures.

Chat2SQL Code | *Java, TypeScript, Snowflake, SQL Server, Oracle, Docker*

- Extended an AI-assisted SQL workspace on Java and TypeScript supporting 10+ databases, including **Snowflake**, SQL Server, and Oracle, generating schema-aware queries with 82% accuracy and validated execution.

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP, REST APIs*

- Engineered a cloud data ingestion and indexing pipeline over 50K+ documents with **incremental updates**, metadata persistence in PostgreSQL, and FAISS retrieval for LLaMA-2 Q&A, reducing retrieval latency 40%.